

ENGINEERING THESIS

# Smart Examination Seating Arrangement System

SUBMITTED BY

**M. Shravya**

Roll No: 2520030588

Dept: CSE | KLH University

g.Keerthi

2520030416

GUIDED BY

**Dr. G. Sudhakar**

Designation: Professor

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# CONTENTS & AGENDA



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## Proposal

Abstract of the seating system solution.



## Requirements

Technical hardware and software stack.



## Analysis

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## Architecture

Logic flow and module structures.



## Final Review

Advantages, Conclusion, and Q&A.

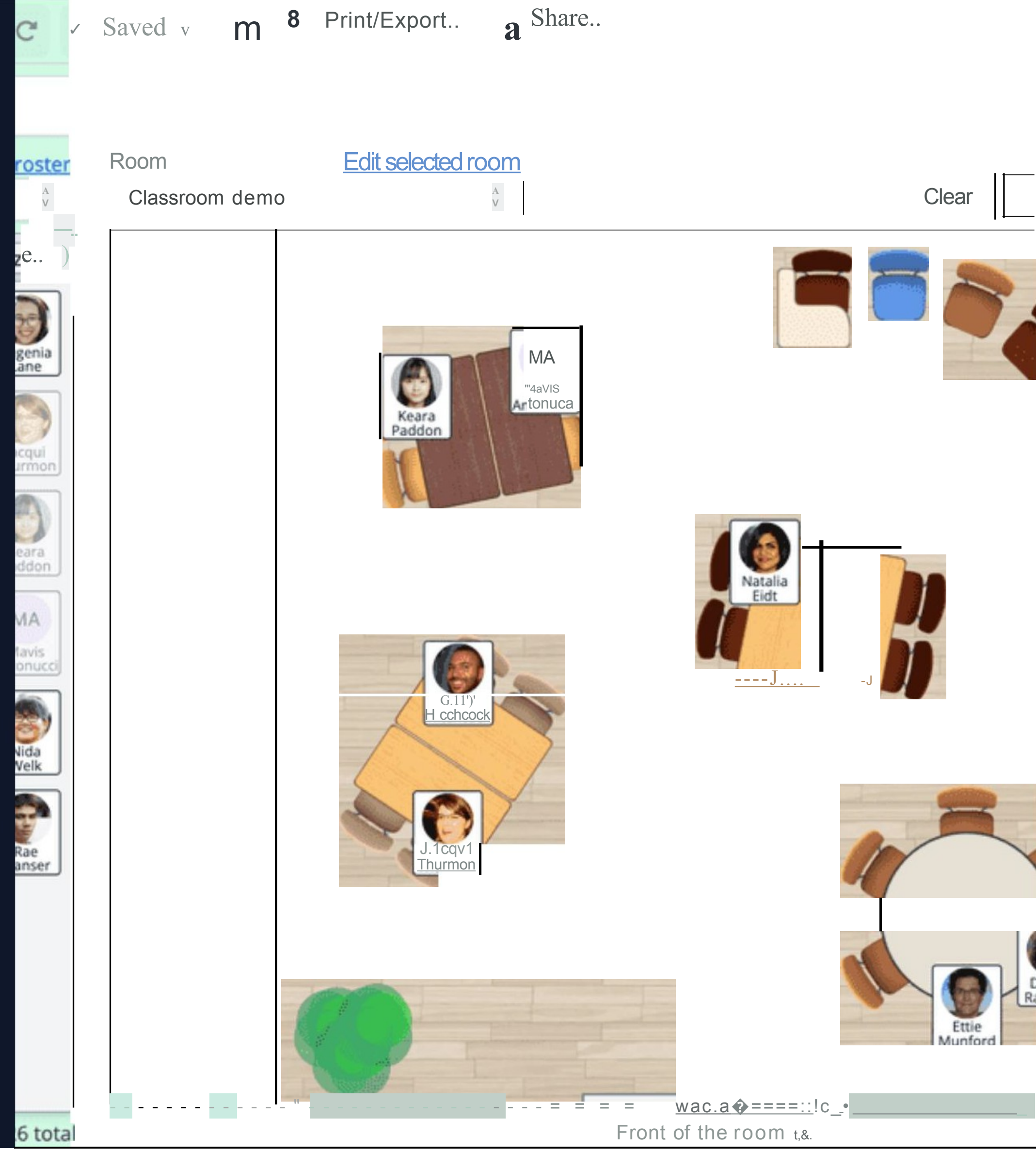


# ABSTRACT & PROPOSAL

The Smart Examination Seating Arrangement System is a software application designed to **automate** the process of assigning seats to students during examinations.

## Key Objectives:

- ✓ Automate seat allocation using student and classroom data.
- ✓ Ensure fair distribution and prevent malpractice.
- ✓ Maximize examination hall resource efficiency.
- ✓ Reduce administrative workload and human errors.



# HARDWARE REQUIREMENTS

|           |                        |
|-----------|------------------------|
| Processor | Intel Core i3 or above |
| RAM       | 4 GB or higher         |
| Hard Disk | 500 GB HDD/SSD         |
| Monitor   | 15-inch HD Display     |
| Input     | Keyboard & Mouse       |
| Output    | Laser Printer (Charts) |





# SOFTWARE REQUIREMENTS



## Operating System

Windows 10 / 11 (64-bit)



## Frontend

HTML, CSS, JavaScript



## Backend

Python / PHP Frameworks



## Database

MySQL / SQLite



## IDE / Tools

VS Code / PyCharm



## Web Browser

Google Chrome / Edge

# | EXISTING MANUAL SYSTEM

## Process

- ✓ Manual preparation of charts.
- ✓ High human effort & time.
- ✓ Prone to duplication errors.
- ✓ Rigid to last-minute changes.

## Limitations

- ✓ Poor resource utilization.
- ✓ High administrative cost.
- ✓ Difficulty in scaling for students.
- ✓ Security & malpractice risks.

# EXISTING SYSTEM ARCHITECTURE





# PROPOSED SMART SYSTEM

## Automated Intelligent Logic

Our solution utilizes algorithms to instantly generate seating plans based on hall capacity and student strength, ensuring zero overlaps.

- ✓ Instant Generation
- ✓ Optimal Spacing
- ✓ Malpractice Prevention
- ✓ Digital Archives





# PROPOSED SYSTEM ARCHITECTURE



# SYSTEM ADVANTAGES



## Efficiency

Reduces calculation time from hours to seconds.  
Drastically minimizes manual data entry effort.



## High Accuracy

Eliminates human errors such as double-seating or  
incorrect roll number mapping.



## Security

Prevents conflicts and ensures students are placed  
in a way that minimizes cheating opportunities.

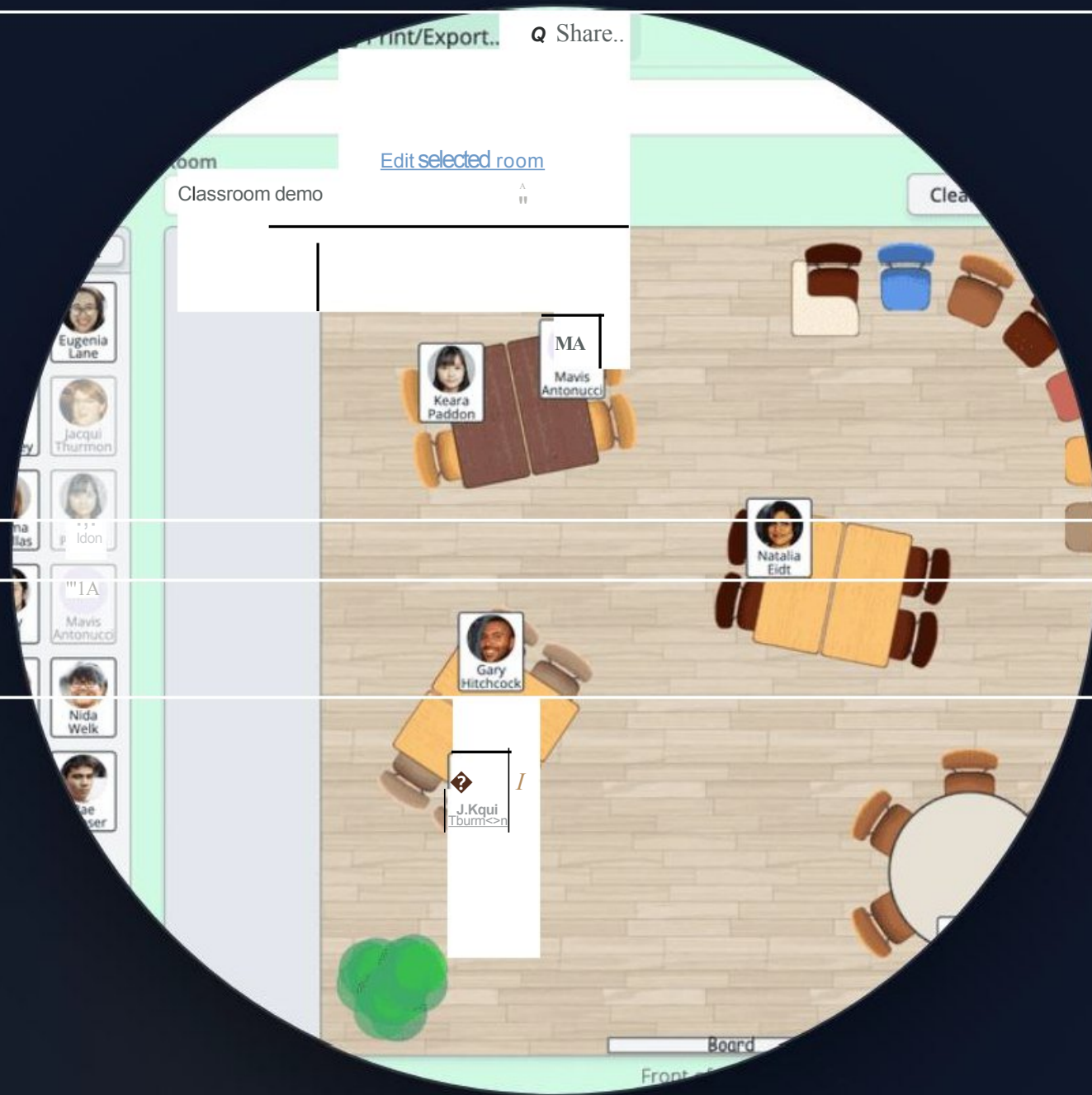


## Resource Scaling

Easily handles large datasets of students and multi-  
building examination complexes.



# CONCLUSION



The Smart Examination Seating Arrangement System provides a **robust, efficient, and scalable** solution for modern educational institutions.

By automating the seat allocation process, the system helps institutions conduct examinations in a more organized, secure, and error-free manner, paving the way for digital administration.

# Thank You

## Questions & Discussion

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Presented by M. Shravya

CSE Department | KLH University



# IMAGE SOURCES



<https://seatingchartmaker.app/static/4581525a8ca10384f130c1e4d5fb7159/52ca5/classroom-hero-04-24.png>

Source: [seatingchartmaker.app](https://seatingchartmaker.app)

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Source: [pngtree.com](https://png.pngtree.com)

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